

MATH120-24S2 Discrete Mathematics

Tutorial 9

References

- Topics 15 and 16: Graphs and Trees
- Sections 10.1, 10.2, 10.5 and 10.7 of Epp (4th Ed.). Sections 10.1, 10.4 and 10.6 of Epp (5th Ed.)

Work on the following problems in groups during the tutorial in week 10.

- ⇒ 1. [B] In a group of 31 people, is it possible that everyone shakes hands with exactly 3 other people? Explain.

Solution:

We can interpret this as a graph problem. In particular, is there a 31-vertex graph G where each vertex has degree exactly three? If so, then

$$\sum_{v \in V(G)} d(v) = 31 \times 3 = 93.$$

But, by the Handshaking Lemma, this sum is twice the number of edges. Hence G has $\frac{93}{2} = 46\frac{1}{2}$ edges; a contradiction. Hence it is not possible.

- ⇒ 2. [B] Show that in a bipartite graph, every cycle has an even number of edges.

Solution:

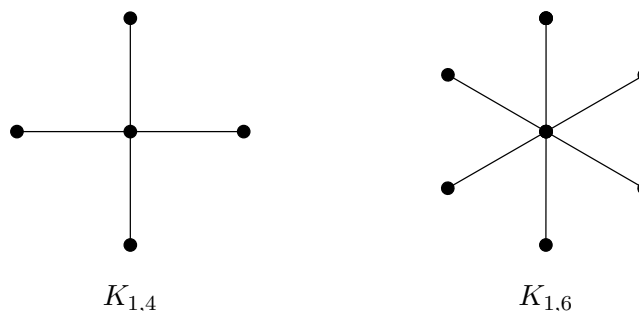
Let G be a bipartite graph with vertex sets V_1 and V_2 . Let C be a cycle in G . Then every edge in C joins two vertices of which one is in V_1 and the other in V_2 . That means that the cycle alternates between vertices in V_1 and V_2 . Since in a cycle, the first and last vertex coincide, we find that the number of different vertices, and hence, the number of edges in C is an even number.

3. [D/M] For all $n \geq 1$, the complete bipartite graphs $K_{1,n}$ are called *star graphs*.

- (a) Draw $K_{1,4}$ and $K_{1,6}$.
 (b) Prove that the star graphs are the only complete bipartite graphs that are trees.

Solution:

(a)



- (b) First we show that every star graph is a tree. Let $\{V_1, V_2\}$ be the bipartition of the vertex set of $K_{1,n}$, where $|V_1| = 1$, $|V_2| = n$, and no edge joins two vertices in V_2 . Since there is a vertex in $K_{1,n}$ joined to every other vertex, $K_{1,n}$ is connected. Furthermore, $K_{1,n}$ contains no cycles, as $K_{1,n}$ is simple and so any cycle would contain at least two distinct vertices in V_1 ; a contradiction.

Now we show that the only complete bipartite graphs that are trees are the star graphs. We will show the contrapositive of this statement. So suppose $K_{m,n}$ is a complete bipartite graph, where $m \geq 2$ and $n \geq 2$, that is, $K_{m,n}$ is not a star graph. Let $\{U, V\}$ be the bipartition of the vertex set of $K_{m,n}$, where no edge joins two vertices in U and no edge joins two vertices in V . Let $u_1, u_2 \in U$ and let $v_1, v_2 \in V$. Then, as every vertex in U is joined by an edge to every vertex in V , the bipartite graph $K_{m,n}$ has a cycle

$$u_1 v_1 u_2 v_2 u_1.$$

It follows that $K_{m,n}$ is not a tree.

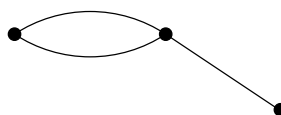
$\Rightarrow$ 4. [M-C] Recall that a graph is *simple* if it has no loops or parallel edges.

- (a) Let G be a simple connected graph with at least two vertices. Show that G has two vertices of equal degree.

- (b) Show that if G is not simple, then G may not have two vertices of equal degree.

Solution:

- (a) Let n be the number of vertices in G . Since G is connected, every vertex has at least degree 1 and since G is simple, the maximum degree is $n - 1$. So the degree of each vertex is a number in $\{1, \dots, n-1\}$. Since there are n vertices with only $n-1$ possibilities for their degrees, at least two vertices have the same degree (this is the pigeonhole principle).
- (b) In the non-simple graph below, no two vertices have equal degree.



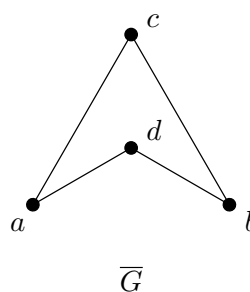
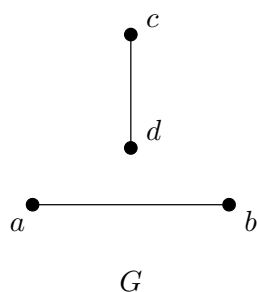
$\Rightarrow$

5. Let G be a simple graph on n vertices. IF G is a forest, how many edges does it have? Justify your answer.

Solution:

A forest with k components has $n - k$ edges. This is because it consists of k trees, each on n_i vertices and $n_i - 1$ edges where $\sum_{i=1}^k (n_i) = n$. So the sum of edges is $\sum_{i=1}^k (n_i - 1) = n - k$.

6. [M] The *complement* $\overline{G}$ of a simple graph G is a simple graph with the same vertex set as G , and has an edge joining two vertices u and v precisely if no edge joins u and v in G . Prove that for any simple graph G , either G or $\overline{G}$ is connected.



(Hint: Assume that G is disconnected and show that $\overline{G}$ is connected.)

Solution:

If G is connected, then we are done. Therefore assume that G is disconnected. Let u and v be vertices of G . We need to show that u and v are connected in $\overline{G}$. We consider two cases: (i) u and v are in different components of G and (ii) u and v are in the same component of G .

- (i) As u and v are in different components of G , no edge joins u and v in G . Therefore, in $\overline{G}$, there is such an edge. Thus u is connected to v in $\overline{G}$.
- (ii) Since G is disconnected, there is a component C of G whose vertex set does not contain either u or v . Choose a vertex w of C . Now no edge of G joins u and w or joins v and w . Therefore, in $\overline{G}$, there is an edge that joins u and w , and an edge that joins w and v . Hence, u is connected to v in $\overline{G}$.

7. [M] Prove that a tree on at least two vertices must contain at least two vertices of degree 1.

Solution:

A tree with n vertices has $n - 1$ edges. The handshaking lemma says that $2|E| = \sum_{v \in V(G)} (d(v))$.

If all n vertices have degree at least 2, then $2|E| = 2(n - 1) = \sum_{v \in V(G)} (d(v)) \geq 2n$, a contradiction. So we find at least one vertex with degree 1.

If we find exactly one such vertex however, we have $2|E| = 2(n - 1) = \sum_{v \in V(G)} (d(v)) \geq 1 + 2(n - 1)$ which is again a contradiction.

Therefore, there are at least 2 vertices of degree 1.

8. [C] Let G be a connected graph in which all vertices have even degree. Show that no edge of G is a bridge (isthmus).

Hint: Use the Handshaking Lemma.

Solution:

Let $e = \{u, v\}$ be an edge of G . Suppose that e is a bridge of G . Then, by Lemma 16.3, the graph obtained from G by removing e has

exactly two components. One component C_u contains u and the other component C_v contains v . Since every vertex of G has even degree, every vertex in C_u has even degree except for u which has odd degree. But C_u is a connected graph and so, by Corollary 15.2, it has even number of vertices of odd degree; a contradiction. Thus no edge of G is a bridge.